ASSIGNMENT # 3 – TRAINING AN LLM

Please see my **GitHub** for Assignment #3:

https://github.com/smurphyphd/text-analysis-public-university-llm/tree/main

Part I:

1. **Think of a dataset (corpus) and a classification task. Ideally, both the corpus and the classification task can be used in your final paper. However, it's OK if you do this for the assignment (you will still need a corpus). You can choose any task except sentiment classification.**

The GitHub folder Assignment 3 contains the excel file I used to build my dataset and classify the data myself. This excel file, called *"trainingdataset_ontario_budgets_v4_llm_sample".xlsx*, contains 202 observations. Each observation represents a paragraph from the introduction section of the operating budget of an Ontario university in advance of an academic year. My current sample contains operating budgets from Queen's University and University of Toronto between 2017 and 2023. I selected this as an initial sample as Queen's and UofT at times have had opposing financial results and leadership approaches to market conditions.

In 2019, the Ontario government-imposed restrictions on domestic tuition fees, which some Ontario universities have attributed as the cause of poor financial performance post-2019. I am interested in how publicly funded universities in Canada talk about how dependent they are upon their provincial government. I want to see if there are differences in how these organizations perceive the external market and their place within it and if those differences are correlated with differences in financial results incurred t+1. I am choosing to analyze the introduction section of the operating budget of each Ontario university, created in advance of the academic school years pre and post 2019. These introduction sections, sometimes called "setting the context" or "the changing financial landscape", outline the university's perspective on its market conditions, the policy decisions of its provincial and national government, and how financially vulnerable or adaptable is their institution.

2. **Decide the number of categories that you will be predicting.**

Academic Definition: Resource Dependence

Emerson (1962) argued that power resides implicitly in one's resource dependence on the other in a relationship. The dependence of actor A upon actor B is (1) directly proportional to A's motivational investment in goals mediated by B, and (2) inversely proportional to the availability of those goals to A outside of the A-B relation. Goals = gratifications consciously sought as well as rewards unconsciously obtained through the relationship**.** Availability = alternative avenues of goal-achievement, most notably other social relations.

Emerson, R. 1962. “Power dependence relations.” *American Sociological Review* 27: 31-40.

I decided to code 2 categories (dependence = 1, non-dependence = 0). I am coding the university’s dependence on their provincial government for resources, as determined by their own perception or projection of that dependence in writing. I initially considered building 3 categories (low, medium and high dependence), but given the size of my sample for this assignment and this being my first time using LLMs to annotate and finetune, practicing with a binary category was a simpler initial approach. I may use a 3 category approach for my final assignment.

3. **Decide the number of observations you will code per category.**

The following represents the breakdown of low vs high dependence observations I have in my sample after hand coding (see V4 of excel).

TOTALS

LOW	HIGH	TOTAL
82	120	202

4. **Create a draft codebook to guide coders who will (hypothetically) label your training set.**

This codebook is related to the classifications I made in step 4.

Low Dependence:

Indicates a university’s low concern for the rewards/financial resources mediated by the provincial government (Emerson)

Quotes:
- *the need to diversify revenue remains pressing*
- *promotes the vision of the university, transformative, the guiding policies*
- *differentiation envelope tied to performance in priority areas, this was a welcome change for the university and reflected the university's long-term advocacy for differentiation*
- *the federal budget included significant new investments in research, new investments in canada research chairs, chairs make an important contribution to*

the university's operating budget, have a significant impact on the university's ability to recruit and retain outstanding scholars

- *the university faces increasing financial pressure as a result of constrained provincial tuition and enrolment frameworks and real value decreases in provincial operating grants, the university's commitment to being an internationally significant research university requires creative solutions to fund its mission and aspirations*

AND/OR

Indicates a university has a large perception of the availability of rewards/financial resources outside of the traditional relationship with the provincial government (Emerson)

Quotes:
- *flexibility, contingency fund*
- *funding from the federal government is not generally part of the university's operating budget, however, it interacts with the University's budget in three important areas: canada research chairs, funding for the indirect costs of research, graduate student support*
- *it is anticipated a greater portion of future funding will be directed towards achievement of accountability metrics, engaged in consultation with the province on the design of metrics and funding mechanisms that will be included in SMA3*
- *the final report of the alternative funding sources advisory group articulated several potential sources of revenue-generation that take advantage of some of the university's key strengths, underpinned by overarching principles that are committed to protecting the university's reputation, building a pipeline of new ideas, financial flexibility*
- *the university has seized another such opportunity with the adoption of the four corners strategy, leverage real estate assets, support the academic mission, simultaneously grow revenue from sources other than enrolment, sets an ambitious goal of generating $50 million in operating funding per year by 2033, development of new space, funding invested in research and teaching mission, several projects are now underway in various stages of planning, design and construction, this new revenue stream is not yet reflected in long-range budget assumptions*

High Dependence

Indicates a university's strong concern for the rewards/financial resources mediated by the provincial government (Emerson)

Quotes:
- *almost 97% of operating grants*
- *found itself grappling, the effects not yet fully understood*

- *funding in the performance-based envelope will be at-risk unless the targets are continually met*
- *provincial grant revenue is financially limiting, any sustained growth above the mid-point does not result in additional provincial grant funding*
- *the provincial government announced the tuition framework, which continues to restrict flexibility for the university by freezing tuition fees*

AND/OR

Indicates a university has a limited perception of the availability of rewards/financial resources outside of the traditional relationship with the provincial government (Emerson)

Quotes:
- *limited flexibility, much of this revenue is directed and regulated*
- *measures and requirements were expected to impact university*
- *however the province will provide institutions with the ability to increase tuition fees for domestic out-of-province students*
- *all units planned their budgets and reductions, the funding that was cut was held in a deficit mitigation fund, the entire deficit mitigation fund allocated to FAS along with a portion of the transition fund (university fund)*
- *the ministry re-introduced the funding linked to SMA3 metrics but at lower rates than were originally contemplated, this now places 10% of funding at risk for 2023-2024 and 25% at-risk for 2024-2025, the budget does allow for some revenue reduction*

This is the version of that codebook I put into Google Colab for the LLM annotation:

COLAB CODEBOOK

```
prompt = """
Please classify the following sentence as either:
- 1 = resource dependence language
- 0 = non-resource dependence language

Resource dependence language refers to situations where an organization is
constrained by, dependent on, or limited by external actors (e.g., government
policies, funding structures, regulations).

This includes:
- funding being at risk or uncertain
- restricted flexibility due to external rules
- dependence on provincial or external funding decisions
```

- externally imposed constraints on operations

Only assign 1 if the sentence clearly reflects external constraint or dependence.

Examples of 1 cases:

the ministry re-introduced the funding linked to SMA3 metrics at lower rates, placing a portion of funding at risk

the provincial government announced a tuition framework that restricts flexibility by freezing tuition fees

much of this revenue is directed and regulated by the provincial government

Examples of 0 cases:

the university is pursuing alternative sources of revenue and diversification strategies

the university launched a strategy to generate new revenue streams from real estate assets

faculty emphasized innovation, differentiation, and long-term growth

Return only one character: 0 or 1.
""".strip()

5. **Label a sample of your data (N=200); decide how you will sample the data and explain your decision. Have a LLM label the same sample (you can find the code here). Estimate inter-coder reliability and evaluate the results.**

See above. Data was human coded as 82 low and 120 high = 202 total observations.

I made a copy of the Google Colab coded provided in class to use LLM as my annotator: https://colab.research.google.com/drive/1EZQ2k3D6pcOc5l1rMuQgaX5Sxy0Eap6B?usp=sharing

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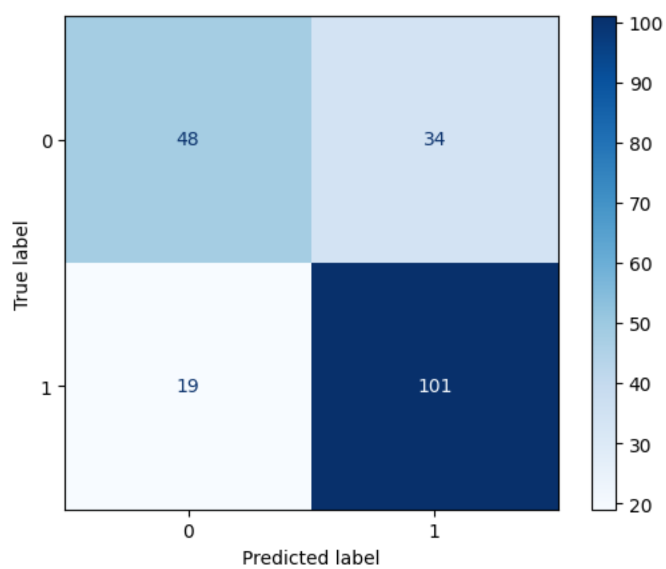
The Zero-Shot, Memory and Memory + Feedback models all provided similar results (see excel file on Github, *"llm_all_predictions".xlsx)*. See screenshots results below from Google Colab:

	method	accuracy	f1_weighted	f1_label_0	f1_label_1	n_valid_predictions
2	zero-shot / one-shot	0.738	0.732	0.644	0.792	202
1	memory + feedback	0.733	0.727	0.635	0.789	202
0	memory	0.723	0.717	0.627	0.780	202

Zero-Shot




```
Zero-shot / one-shot: 100%|██████████| 202/202 [01:43<00:00,  1.96it/s]Zero-shot / one-shot annotation
Accuracy: 0.738
F1 (label 0): 0.644
F1 (label 1): 0.792
F1 (overall): 0.732

              precision    recall  f1-score   support

           0      0.716     0.585     0.644        82
           1      0.748     0.842     0.792       120

    accuracy                          0.738       202
   macro avg      0.732     0.714     0.718       202
weighted avg      0.735     0.738     0.732       202
```

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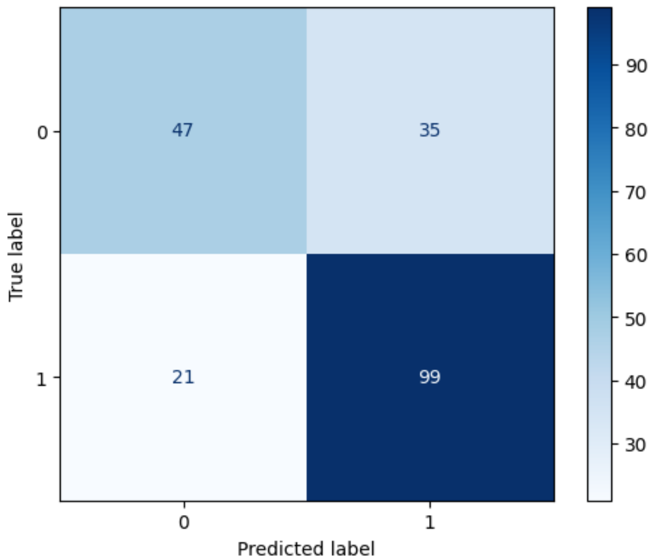
Memory



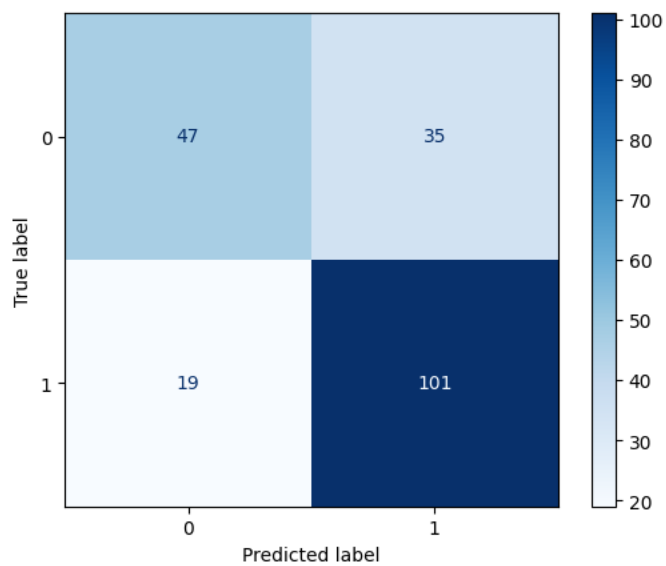

```
Memory: 100%|██████████| 202/202 [02:45<00:00,  1.22it/s]Memory-based annotation
Accuracy: 0.723
F1 (label 0): 0.627
F1 (label 1): 0.78
F1 (overall): 0.717

              precision    recall  f1-score   support

           0      0.691     0.573     0.627        82
           1      0.739     0.825     0.780       120

    accuracy                          0.723       202
   macro avg      0.715     0.699     0.703       202
weighted avg      0.719     0.723     0.717       202
```

Memory + Feedback




```
Memory + feedback: 100%|██████████| 202/202 [06:07<00:00,  1.82s/it]Memory with feedback
Accuracy: 0.733
F1 (label 0): 0.635
F1 (label 1): 0.789
F1 (overall): 0.727

              precision    recall  f1-score   support

           0      0.712     0.573     0.635        82
           1      0.743     0.842     0.789       120

    accuracy                          0.733       202
   macro avg      0.727     0.707     0.712       202
weighted avg      0.730     0.733     0.727       202
```

Q5: Intercoder Reliability

See RMarkdown in GitHub (Assigment 3 – ICR. RMarkdown).

I calculated intercoder reliability in R, comparing my human classification to the three different LLM models. Overall intercoder reliability appears moderate (Fleiss' κ = 0.588). However, it is the pairwise agreement amongst LLM models that is high (κ = 0.72–0.82), whereas agreement between the human coding and each LLM is substantially lower (κ ≈ 0.41–0.44). This suggests that the LLMs are internally consistent but only partially capture the human judgment I did with my coding.

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Q5: Evaluating the Results

The biggest discrepancy I noticed in the results was the difficulty in predicting 0s versus predicting 1s. While the accuracy rate on predicting "dependence" was around 84% (101/120), the accuracy rate on predicting "no dependence" averaged closer to 57% (47/82). In other words, the LLM models (relying upon my codebook) are generating significant false positives (35, Type I error) but relatively fewer false negatives (19, Type II error). The LLMS models are ultimately overpredicting dependence language in the universities when compared to my own human coding.

document	ID	text	class	order	total para	university	province	year	atment gro	atment per	res_externa	pres_exp	rov_extern	prov_exp	pres_new	prov_new	zero_shot_pred	memory_pred	memory_feedback_pred	Variance_zero	variance_memory	variance_memory_feedback
queens_20	queens_20	As part of S	1	3	8	Queen's Un	ON	2025	1	1							1	1	1	0	0	0
queens_20	queens_20	The ministr	1	4	8	Queen's Un	ON	2025	1	1							1	1	1	0	0	0
queens_20	queens_20	On January	0	5	8	Queen's Un	ON	2025	1	1							1	1	1	-1	-1	-1
queens_20	queens_20	For the thre	1	6	8	Queen's Un	ON	2025	1	1							0	0	0	1	1	1
queens_20	queens_20	On Februar	1	7	8	Queen's Un	ON	2025	1	1							1	1	1	0	0	0
queens_20	queens_20	Over the su	1	8	8	Queen's Un	ON	2025	1	1							1	0	0	0	1	1
uoft_2017	uoft_2017_	Canada, an	1	1	21	University o	ON	2017	1	0							0	0	0	1	1	1
uoft_2017	uoft_2017_	The 2016 y	1	2	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	The Provinc	0	3	21	University o	ON	2017	1	0							1	1	1	-1	-1	-1
uoft_2017	uoft_2017_	The Provinc	1	4	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	In 2014 the	1	5	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	The Provinc	1	6	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	The govern	1	7	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	Transition t	0	8	21	University o	ON	2017	1	0							1	1	1	-1	-1	-1
uoft_2017	uoft_2017_	For the 201	0	9	21	University o	ON	2017	1	0							0	1	0	0	-1	0
uoft_2017	uoft_2017_	The Univers	0	10	21	University o	ON	2017	1	0							0	0	0	0	0	0
uoft_2017	uoft_2017_	The Univers	0	11	21	University o	ON	2017	1	0							1	1	1	-1	-1	-1
uoft_2017	uoft_2017_	A full discu	0	12	21	University o	ON	2017	1	0							0	0	0	0	0	0
uoft_2017	uoft_2017_	Tuition fees	1	13	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	The Univers	0	14	21	University o	ON	2017	1	0							0	0	0	0	0	0
uoft_2017	uoft_2017_	At the Univ	0	15	21	University o	ON	2017	1	0							0	0	0	0	0	0
uoft_2017	uoft_2017_	The 2016 O	1	16	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	Funding fro	0	17	21	University o	ON	2017	1	0							1	0	0	-1	0	0
uoft_2017	uoft_2017_	In the 2014	0	18	21	University o	ON	2017	1	0							1	0	0	-1	0	0
uoft_2017	uoft_2017_	The Canada	0	19	21	University o	ON	2017	1	0							1	1	1	-1	-1	-1
uoft_2017	uoft_2017_	As a long-st	1	20	21	University o	ON	2017	1	0							1	1	1	0	0	0
uoft_2017	uoft_2017_	The federal	1	21	21	University o	ON	2017	1	0							0	1	1	1	0	0
uoft_2018	uoft_2018_	The Canadi	1	1	25	University o	ON	2018	1	0							0	0	0	1	1	1
uoft_2018	uoft_2018_	The Bank of	0	2	25	University o	ON	2018	1	0							0	0	0	0	0	0
uoft_2018	uoft_2018_	Ontario's e	1	3	25	University o	ON	2018	1	0							1	1	1	0	0	0
uoft_2018	uoft_2018_	The Provinc	1	4	25	University o	ON	2018	1	0							1	1	1	0	0	0
uoft_2018	uoft_2018_	In 2013, th	0	5	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	Effective 20	0	6	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	Transition t	0	7	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	In October	0	8	25	University o	ON	2018	1	0							0	1	0	0	-1	0
uoft_2018	uoft_2018_	The SMA2 e	0	9	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	The agreem	0	10	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	In addition	0	11	25	University o	ON	2018	1	0							1	1	1	-1	-1	-1
uoft_2018	uoft_2018_	Fall 2017 u	1	12	25	University o	ON	2018	1	0							1	1	0	0	0	1
uoft_2018	uoft_2018_	Plans call fo	0	13	25	University o	ON	2018	1	0							0	1	1	0	-1	-1
uoft_2018	uoft_2018_	The Univers	0	14	25	University o	ON	2018	1	0							0	0	0	0	0	0
uoft_2018	uoft_2018_	The SMA2 e	0	15	25	University o	ON	2018	1	0							0	1	1	0	-1	-1
uoft_2018	uoft_2018_	Tuition fees	1	16	25	University o	ON	2018	1	0							1	1	1	0	0	0
uoft_2018	uoft_2018_	The Univers	0	17	25	University o	ON	2018	1	0							0	0	0	0	0	0
uoft_2018	uoft_2018_	At the Univ	0	18	25	University o	ON	2018	1	0							0	0	0	0	0	0
uoft_2018	uoft_2018_	The propor	0	19	25	University o	ON	2018	1	0							0	0	0	0	0	0
uoft_2018	uoft_2018_	long range	1	20	25	University o	ON	2018	1	0							1	1	1	0	0	0
uoft_2018	uoft_2018_	Funding fro	0	21	25	University o	ON	2018	1	0							1	0	0	-1	0	0
uoft_2018	uoft_2018_	long range	1	22	25	University o	ON	2018	1	0							1	1	1	0	0	0

6. **How difficult or easy was the task? What problems did you run into? What would you change in your codebook to improve it? What other lessons did you learn from this exercise?**

Relatively speaking, this was an easy task. The Google Colab code ran well once I adjusted the API key and inserted my datafile. I did have to write code at the end to export the LLM predictions a separate excel file, because I wanted to have this file on hand.

Looking at the discrepancies between the LLMs and the human coding (as seen in llm_all_predictions.xlsx – highlighted in red), many of them occurred with the University of Toronto sample. Some paragraphs that I coded as being low in dependence, since they reference the government (often the federal government as an alternative funding source) are labelled as high in dependence by the LLMs. I think I need to re-write my codebook to be clearer that dependence language does not necessarily always involve a discussion about the government. Low dependence has more to do about creativity and ingenuity in acquiring funds beyond typical grants or removing the domestic tuition freeze, which may involve some layer of government or be completely independent of government.

I also wonder if - because I gave criteria for 0s and for 1s in my codebook - that added complication for the LLMs. I should be clearer in specifying 1s only, and everything else will be coded as a 0. This also may speak to while using 3 categories (low, medium and high) may be better. I will go through each discrepancy and revise my codebook accordingly.

Part II:

1. **Once you have validated your training set and the labels produced by LLMs, expand the training set until you have (at least) 150 observations per category. You can use LLMs to label, but you will still need to manually validate the labels produced.**

Extra Observations (150 per category)

I expanded my observations so there over 150 observations per category. This now saved as trainingdata_ontario_budgets_V5_llm_sample.xlsx in the GitHub. These observations were all hand-coded.

PART 2		
LOW	HIGH	TOTAL
165	163	328

LLM Validation: Second Attempt

I also updated my codebook in Google Colab and had the LLMs try again. I removed reference to coding 0s and kept it only to coding 1s. I also refined how I define resource dependence for it to be more nuanced that is not strictly relate to referencing the government alone.

Produced the llm_predictions-5 excel file, which is used to calculate intercoder reliability again. See Rmarkdown file for more info.

Revised Codebook:

prompt = """
Please classify the following sentence as either:
- 1 = resource dependence language
- 0 = non-resource dependence language

Resource dependence language refers to situations where a university feels constrained by, dependent on, the government. In particular, resource dependent universities feel limited by Ontario policies, such as the domestic tuition freeze, and talk as though there are not alternative ways for their operating revenues to grow or operating expenses to be covered. Less dependent universities look for alternative funding through their own means but also through looking to different layers of government or different programs outside of the tuition framework.

This includes:
- provincial funding being at risk or uncertain
- restricted flexibility due to domestic tuition freeze
- dependence on provincial government funding decisions
- externally imposed constraints on operations

Only assign 1 if the sentence clearly reflects external constraint or dependence.

Examples of 1 cases:

the ministry re-introduced the funding linked to SMA3 metrics at lower rates, placing a portion of funding at risk

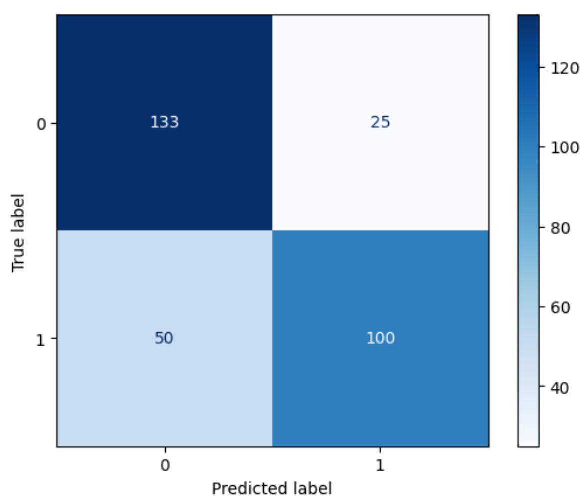
the provincial government announced a tuition framework that restricts flexibility by freezing tuition fees

much of this revenue is directed and regulated by the provincial government

""".strip()

LLM Improvements: V5 excel

Zero Shot



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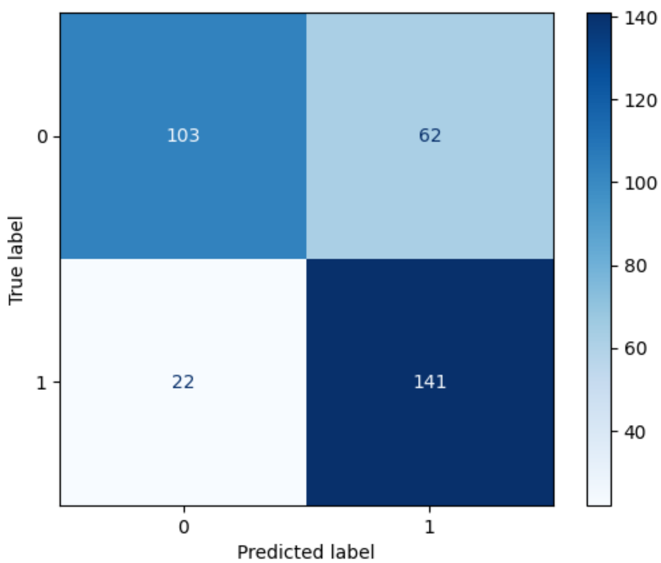
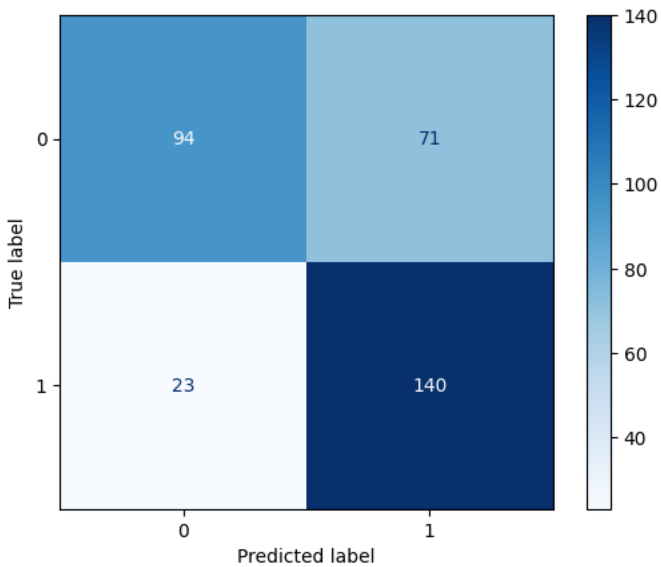
```
Zero-shot / one-shot: 100%|██████████| 328/328 [02:55<00:00,  1.86it/s]Zero-shot / one-shot annotation
Accuracy: 0.756
F1 (label 0): 0.78
F1 (label 1): 0.727
F1 (overall): 0.754

              precision    recall  f1-score   support

           0      0.727     0.842     0.780       158
           1      0.800     0.667     0.727       150

    accuracy                          0.756       308
   macro avg      0.763     0.754     0.754       308
weighted avg      0.762     0.756     0.754       308
```

Memory



	precision	recall	f1-score	support
0	0.824	0.624	0.710	165
1	0.695	0.865	0.770	163
accuracy			0.744	328
macro avg	0.759	0.745	0.740	328
weighted avg	0.760	0.744	0.740	328

Memory Feedback




```
Memory + feedback: 100%|██████████| 328/328 [11:58<00:00,  2.19s/it]Memory with feedback
Accuracy: 0.713
F1 (label 0): 0.667
F1 (label 1): 0.749
F1 (overall): 0.707

              precision    recall  f1-score   support

           0      0.803     0.570     0.667       165
           1      0.664     0.859     0.749       163

    accuracy                          0.713       328
   macro avg      0.733     0.714     0.708       328
weighted avg      0.734     0.713     0.707       328
```

```python
comparison = pd.DataFrame(
    memory_scores + feedback_scores + zero_shot_scores
).sort_values("f1_weighted", ascending=False)

comparison.round(3)
```

	method	accuracy	f1_weighted	f1_label_0	f1_label_1	n_valid_predictions
2	zero-shot / one-shot	0.756	0.754	0.780	0.727	308
0	memory	0.744	0.740	0.710	0.770	328
1	memory + feedback	0.713	0.707	0.667	0.749	328

2. **Using your (validated) training set, fine-tune the encoder-only LLM of your choice.**

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After re-developing my corpus and code, I eventually developed a V14 sample. I used this to fine-tune the encoder. This sample is also used in my Final Assignment. This model predicts no perceived resource dependence versus perceived resource dependence.

See Google Colab link here:

https://colab.research.google.com/drive/1WtSSnQ2e-JwbKbzeNqLQUipELRhO1flC?usp=sharing

3. **Report performance statistics for your fine-tuned model (e.g., accuracy, F1, recall, etc.), as well as the confusion matrix.**

Mean ACC	SD ACC	Mean Weighted FI
0.688	0.077	0.653

Based on best optimal epoch within each fold. Epochs = 4, Folds = 5.

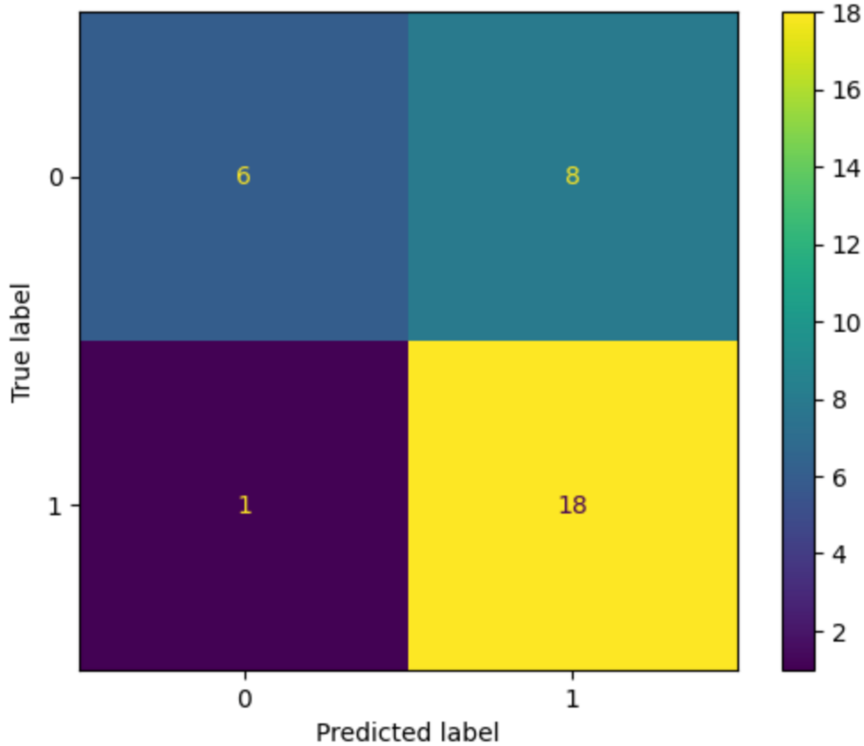
Here is an example of a confusion matrix for a particular fold.

```
True counts: Counter({np.int64(1): 19, np.int64(0): 14})
Pred counts: Counter({np.int64(1): 26, np.int64(0): 7})
              precision    recall  f1-score   support

           0      0.857     0.429     0.571        14
           1      0.692     0.947     0.800        19

    accuracy                          0.727        33
   macro avg      0.775     0.688     0.686        33
weighted avg      0.762     0.727     0.703        33


Confusion Matrix:
[[ 6  8]
 [ 1 18]]
```



Category 1s are being predicted with strong accuracy, but there are 8 category 0s that are incorrectly being predicted as 1s.

4. **Use your trained model to predict labels on a target corpus.**

Using a small sample of Carleton operating budgets (Carleton_sample_predict excel file), found in github.

5. **Describe the results obtained, and whether they match the expectations you had about the data.**

The results match my expectations. The Carleton sample is hand coded with predominately 1s… However, my model is able to effectively identify those as 1s (or as perceived resource dependence. See the confusion matrix below.

```
100%|██████████| 33/33 [00:00<00:00, 49.08it/s]
True labels:
 class_binary

Predicted labels:
 predicted_label
1    31
0     2

Confusion matrix:
 [[ 0  0]
 [ 2 31]]

Classification report:
           precision    recall  f1-score   support

        0     0.000     0.000     0.000         0
        1     1.000     0.939     0.969        33

   accuracy                         0.939        33
  macro avg     0.500     0.470     0.484        33
weighted avg     1.000     0.939     0.969        33
```

The sample data coded as 1s appears to have validity, matching my impressions.

	ID	text
0	carleton_2019_2	There has been a lot of change and transition ...
1	carleton_2019_5	The SMA2 gathers information on system-wide an...
2	carleton_2019_7	Although Carleton is presently in good financi...
3	carleton_2020_4	In parallel, during the Fall of 2019, Carleton...
4	carleton_2020_5	Although Carleton is presently in good financi.